



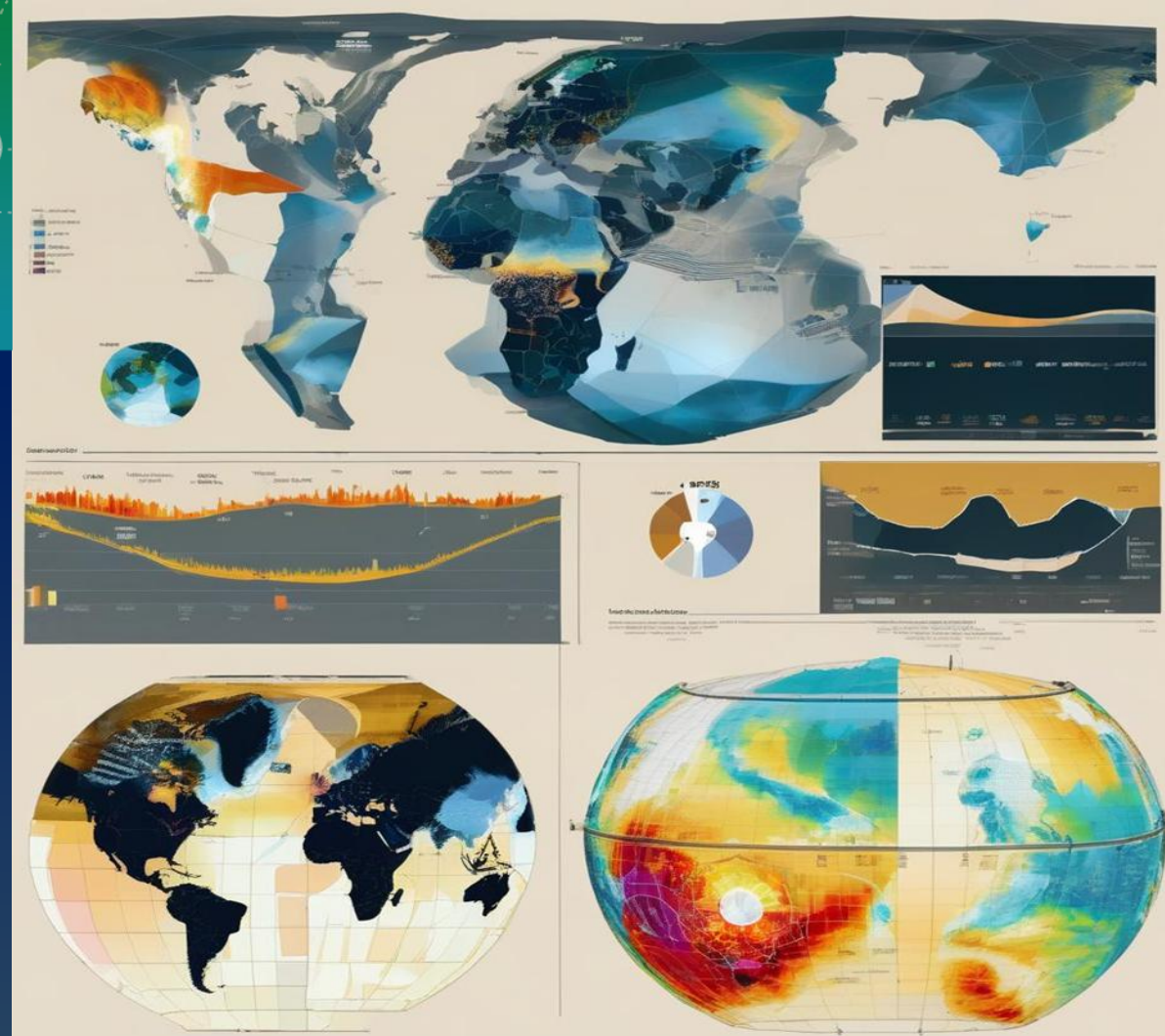
Food and Agriculture
Organization of the
United Nations



Building Capacity for Climate Risk Analysis: Methods, Tools, and Applications for Yemen's Agriculture and Related Sectors

Aden (Yemen), 2 Dec - 4 Dec., 2025

Session I. CIPA framework



Jorge Alvar-Beltrán: jorge.alvarbeltran@fao.org
Natural Resources Officer (Agrometeorology)

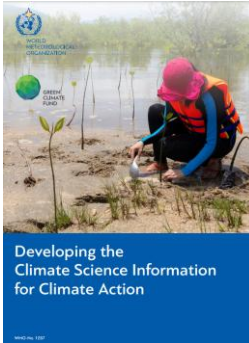
Office of Climate Change, Biodiversity and environment
(OCB, FAO)

Presentation plan : Session I

1. Technical guidance for the formulation of CIPAs

Climate rational

Ensures that the links between climate (hydrological, meteorological and climatological risks) and climate impacts, climate action and benefits to society are fully grounded on the best available climate data and science.



Based on **reliable, relevant and usable climate information**



Evidence for climate investments and policy decision making



Climate impacts assessment on policies and projects



Quality assurance, effectiveness and transformational value of climate action



Source: [WMO](#)

Paradigm shift

This is achieved by providing the most effective mitigation and adaptation options while obtaining the greatest benefits for society.

Mitigation

Climate science for mitigation is fundamental to understanding that reducing greenhouse gas emissions is essential to limiting climate change and its negative impacts.

Adaptation

Climate science for adaptation is essential to selecting actions that reduce exposure/vulnerability to climate change (variability/extreme events), while creating/enhancing opportunities arising from changing climatic conditions.

Importance of impact assessment studies: an impact assessment is one of the six investment criteria areas used by the GCF to evaluate funding proposals. The GCF only funds climate change adaptation and mitigation activities, and therefore, project design must be built around a logic of agricultural interventions that limit climate change risks.

GCF Project Design Cycle: Key Elements

Step 1: Identify the area of intervention

- Consultation process with national stakeholders and prioritization of the targeted sector.

Step 2: Identify relevant contributing climate factors

- Collect historical data, produce future climate projections, and analyze the results.

Step 3: Identify contributing non-climatic factors

- Collect geophysical, natural resource, socio-economic, governance, and infrastructure information.

Step 4: Select effective climate actions

- *Relevance*: The degree to which an action addresses the link between climate hazard and its corresponding impact.
- *Feasibility*: The degree to which an action is practical and achievable under local conditions.
- *Cost-effectiveness*: The degree to which an action demonstrates an impact relative to the economic costs of implementation.

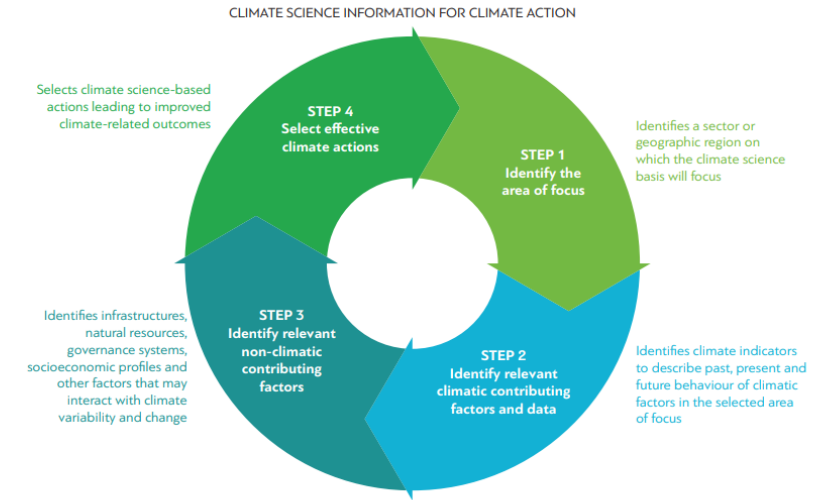


Figure 1. Four-step methodology for developing the climate science information for climate action

Source: GCF

FAO's approach to Climate Impact Potential Assessments (CIPA)

OCB has developed a technical note to guide the development of a Climate Impact Potential Assessment (CIPA) for GCF projects at the Concept Note (CN) and Funding Proposal (FP) stages.

The technical note shows how to articulate climate change challenges and proposed interventions with sufficient information/data to obtain positive feedback from the GCF regarding its investment criteria and assessment areas.

The GCF only funds climate change adaptation and mitigation projects; therefore, project design must be based on an investment logic that balances climate and non-climate contributing factors to climate risks.

Therefore, the selection of project areas should be justified based on emerging findings from climate risk assessments.

Guidance note on Climate Impact Potential for FAO-GCF projects

Version 1 – Updated on 8 March 2024

Summary

This note is intended to provide guidance on how to demonstrate climate impact potential of Green Climate Fund (GCF) projects at GCF Concept Note (CN) stage and the Funding Proposal (FP) stage.

The GCF funds only climate change adaptation and mitigation projects so the project design needs to be constructed around a rationale for investments that modulate climatic and non-climatic contributing factors to climate risks.

It is crucial to understand the agroclimatic context, determine climate risks and impacts and identify interventions that address climate change challenges for targeted areas/sectors by using the best available information. The proposal needs to clearly present linkages between identified climate hazards and impacts, and propose climate change responsive actions that address identified climate risks with highest societal, economic, and environmental benefits.

The selection of project areas should be justified based on emerging findings of the climate risk assessments.

In this guidance note, you will find:

- **What is climate impact potential?** (see section 1.2): The climate impact potential is assessed as the potential contribution of the project's interventions to adaptation and/or mitigation results.
- **How does the GCF assess it?** (section 2): The review of a CN by the GCF focuses firstly on climate impact to determine whether the project is designed to contribute primarily to low-emission and climate resilient development pathways, and whether it can demonstrate, with confidence, that it will provide measurable mitigation and/or adaptation results.
- **What is the GCF guidance on the climate impact potential?** (section 3 and Annex 2): The GCF has established high-level principles for demonstrating the impact potential of GCF-supported activities ([GCF/B.33/05](#) entitled "Steps to enhance the climate rationale of GCF-supported activities").
- **What are the tools/information sources available?** (see Annex 1)
- **Which questions need to be answered for mitigation?** (section 3.23.2 Mitigation):
 - o What is the emissions profile?
 - o What are the regulatory frameworks in place to reduce emissions potential?
 - o What are the mitigation needs?
 - o What are the barriers to implement mitigation activities?
 - o What kind of activities would be implemented to reduce emissions/sequester carbon?
 - o What is the emission reduction potential of the project activities using appropriate methodologies for GHG calculation?

FAO's approach to Climate Impact Potential Assessments (CIPA)

What questions need to be answered for adaptation projects?

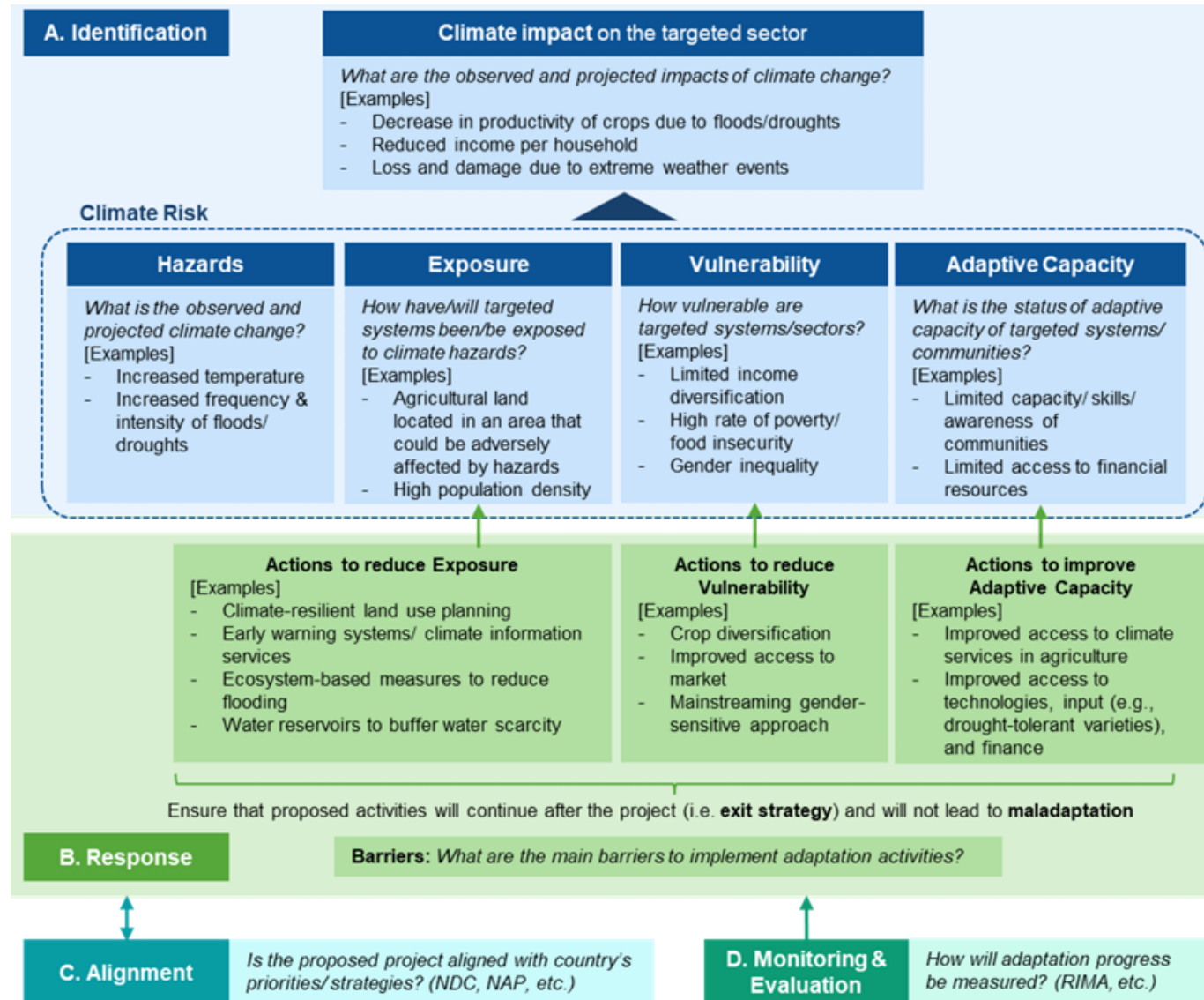
- What are the main climate changes that have occurred in the past?
- How will the climate change in the future?
- What are the observed and projected impacts of climate change?
- How have the targeted systems been exposed to climate risks?
- How vulnerable are the targeted systems/communities to climate change?
- What is the state of their adaptive capacity to cope with climate challenges?
- Which area of the country is the most vulnerable?

FAO's approach to Climate Impact Potential Assessments (CIPA)

What questions need to be answered for adaptation projects?

- Why was the project's target area selected from a climate change perspective?
- What adaptation components does this project aim to implement to address the identified climate risks and barriers hindering adaptation activities?
- Is the proposed project aligned with the country's priorities/strategies?
- What current climate-related strategies/interventions are in place, and how will the proposed GCF project complement and/or address key gaps?
- How will progress on adaptation be measured?
- What indicators will be monitored?

FAO's approach to climate impact potential assessments



FAO's approach to Climate Impact Potential Assessments (CIPA)

Where to find climate data/information?

- To obtain information on observed climate change at the national level, project design teams should contact national institutions - as local climate data are often available from National Meteorological and Hydrological Services.
- Meteorological Services may also have information about future climate projections.
- The National Communication to the UNFCCC and other national climate change reports also present historical and future climate data. The latter can complement the discussion of our findings.
- A wide range of tools and information platforms exist to assist in retrieving and analyzing climate model information.

FAO's approach to Climate Impact Potential Assessments (CIPA)

Non-climate barriers/drivers that can exacerbate climate risks as a vulnerability or adaptive capacity factor include:

- **Conflict:** This is one of the factors that hinder climate change adaptation activities. At the same time, climate threats could amplify conflict risks, particularly in fragile states.
- **Institutional barriers:** Weak institutions are a major cause of poor implementation of climate action. In countries where the role of national institutions is weak, the implementation of such policies by subnational governance levels is limited.
- **Local and technical knowledge:** The technical capacity of extension agents and farmers to adapt to climate change may be insufficient in some contexts and, consequently, hinder the capacity to build climate resilience, including planning, programming, monitoring, and resource mobilization.

FAO's approach to Climate Impact Potential Assessments (CIPA)

Non-climatic barriers/drivers that can exacerbate climate risks as a vulnerability or adaptive capacity factor include:

- **Gender:** In some contexts, female farmers rarely have land ownership rights and often relinquish land rights to male family members, increasing their vulnerability.
- **Social and cultural barriers:** Communities' adaptive capacity is limited due to poor access to the information, markets, services, and technology needed for climate-resilient livelihoods, particularly for women and other vulnerable groups.
- **Financial barriers:** In some decentralized countries, the capacity of subnational institutions to mobilize climate finance and implement adaptation and mitigation efforts is limited.

FAO's approach to Climate Impact Potential Assessments (CIPA)

- Barriers to adaptation are closely linked to the risks of maladaptation due to the interconnections between resources, capacities, and coordination, which can significantly impact adaptation practices and their outcomes.
- Maladaptation, as a concept, generally refers to adaptation initiatives that create additional vulnerabilities instead of reducing them.
- While some initiatives may promote short-term adaptation, there is a risk that they could affect the capacity of territories, sectors, and populations to manage the medium- to long-term impacts of climate change.
- Project proposals should discuss how the proposed activities will not lead to maladaptation and how the risks of maladaptation can be mitigated.

FAO's approach to Climate Impact Potential Assessments (CIPA)

Identification

The first step of an adaptation project is to identify high-risk systems within one or more countries, as well as the climate change and hazards that affect them or could affect them in the future.

Response

For adaptation projects, response actions are planned to reduce negative climate impacts by decreasing exposure to climate change, reducing vulnerability, and strengthening adaptive capacity, thereby increasing resilience.

Resilience focuses on the dynamic capacity of a system to recover from the impacts of climate change and adapt to a changing environment over the long term. Resilience refers not only to shocks (weather extremes) but also to changes in the average itself.

The proposed interventions must be clearly linked to climate risks. Several adaptation interventions can be identified, and they may require prioritization based on a cost-benefit analysis to justify that the proposed interventions constitute the best possible adaptation measures.

FAO's approach to Climate Impact Potential Assessments (CIPA)

ALIGNMENT

Proposals should confirm the alignment of the proposed activity with the country's national climate plans and strategies (including their National Adaptation Plans, long-term climate strategies, including those submitted as Nationally Determined Contributions), as well as with previous/ongoing initiatives/projects and future projects under development.

This helps ensure country ownership of the project proposal and that activities target areas of greatest need and potential impact for that country.

MONITORING AND EVALUATION

Proposals should include a description of the monitoring and evaluation system that will be used to assess the impact of the proposed activity and quantify the beneficiaries of the adaptation.

This will facilitate the evaluation, during implementation, of whether the Theory of Change is working, whether the funded activity has generated the expected result, and how they inform the design of more beneficial future adaptation options.

List of tools and information sources

Guiding questions	Name	Thematic area	Description/remark	Scale	Type of tool/sources
Adaptation					
(a)	Climate Information Platform (CIP)	Climate change trend and projection	The CIP co-developed by the GCF and WMO provides a rapid regional assessment of selected climate indicators.	- National - Site-specific	Platform
(a)	World Bank Climate Change Knowledge platform (CCKP)	Climate change trend and projection	The CCKP contains climate, disaster risk, and socio-economic datasets, as well as synthesis products, such as the Climate Risk Country Profiles.	- National - Sub-national	Platform
(a)	IPCC Assessment Reports (6th Assessment Report , WGI in particular) Interactive IPCC Atlas	Climate change trend and projection	This report and interactive atlas can provide some useful information on observed and projected climate change.	- Global - Regional - National	Report Platform
(e)	Climate Risk Toolbox (CRTB)	Climate risk	The CRTB is hosted by the FAO Hand-in-Hand Geospatial Platform and provide quick climate risk assessment.	- National - Sub-national (not all data)	Platform
(a)	FAO CAVA Platform FAO CAVAnalytics	Climate change trend and projection	CAVA platform provides the most essential climate change information in a user-friendly interface. The more advance CAVA Analytics tool can also be provided upon request (contact OCB Risk Team).	- National - Sub-national	Platform
(b) (c)	Emergency Events Database (EM-DAT)	Climate induced disasters	EM-DAT launched by the Centre for Research on the Epidemiology of Disasters (CRED) contains essential core data on the occurrence and effects of over 22,000 mass disasters in the world from 1900 to the present day.	- National - Sub-national	Database
(b) (c)	DesInventar¹⁴	Climate induced disasters	DesInventar provides data about damage, losses and the effects of disasters.	- National - Sub-national	Database
(b) (c)	WMO	Climate induced disasters	WMO's report provides an overview of mortality and economic losses from weather, climate and water related hazards from 1970 to 2019.	- Global - Regional	Report

¹⁴ [A new disaster losses and damages tracking system](#) will replace the DesInventar once it becomes available.

List of tools and information sources

Guiding questions	Name	Thematic area	Description/remark	Scale	Type of tool/sources
(b)	Global Agro-Ecological Zones (GAEZ) assessment	Historical and projected impacts of climate change	The GAEZ v4 spatial data are organized in six themes: (1) Land and Water Resources, (2) Agro-climatic Resources, (3) Agro-climatic Potential Yield, (4) Suitability and Attainable Yield, (5) Actual Yields and Production, and (6) Yield and Production Gaps	- National - Sub-national	Platform
(b)	Climate Adaptation in Rural Development (CARD) Assessment Tool	Projected impacts of climate change	This tool developed by IFAD can provide information on the effects of climate change on the yield of major crops.	National	Excel-based information
(b) (f)	FAO's report " Managing risks to build climate-smart and resilient agrifood value chains " (2022)	Impacts of weather hazards	This report can give you an idea of possible climate- and weather-related hazards and impacts along the agrifood value chain, besides recognizing the added value of climate services on building climate resilience	Global	Report
(b) (f)	Global Outlook on Climate Services in Agriculture: Investment opportunities to bridge the last-mile gap (2021).	Impacts of weather hazards	This report provides an outlook of the state of weather-informed agricultural advisories worldwide, including regional barriers and opportunities along the climate services value chain	- Global - Regional - National	Report
(d) (i)	FAO's Resilience Index Measurement and Analysis tool (RIMA)	Measurement of adaptation progress	RIMA provides ideas for how to measure resilience in the context of climate change	Household level	Methodology
(d) (i)	FAO's Self-evaluation and Holistic Assessment of Climate Resilience of Farmers and Pastoralists tool (SHARP)	Measurement of adaptation progress	SHARP tool is a holistic assessment of farmers climate resilience at the household level. The improved tool (SHARP+) is available.	Household level	Methodology
Various	AQUASTAT	Statistics		National	Database
Various	FAOSTAT	Statistics		National	Database
Mitigation					
(f)	EX-ACT	GHG emission calculation	EX-ACT is the GHG accounting tool that covers the entire agricultural sector.	Project specific	Methodology
(f)	GLEAM-I	GHG emission calculation	GLEAM is a modelling framework that simulates the interaction of activities and processes involved in livestock production and the environment.	- National - Sub-national	Methodology
(f)	FAOSTAT	Statistics		National	Database

End!
